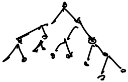
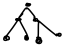


- recursive binary splitting may overfit data with too complex trees
- smaller trees may have lower variance with slightly larger bias
- bad strategy $\Rightarrow$ stop splits early when the entropy reduction is below an high threshold ('better' splits can occur later in the tree)
- good strategy $\Rightarrow$ pruning:
  1. grow a large initial tree T_0 where $|T_0|$ is the number of leaf nodes
  2. for different values of coefficient α determine a sequence of sub-trees $T \subseteq T_0$ minimizing

     HYPERPARAMETER (REGULARIZATION)

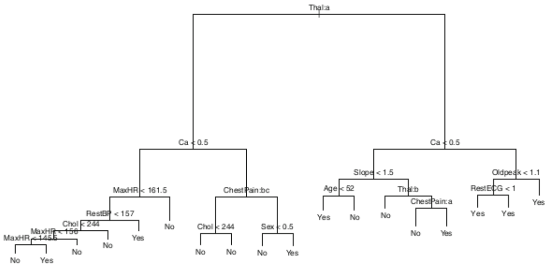
T_0

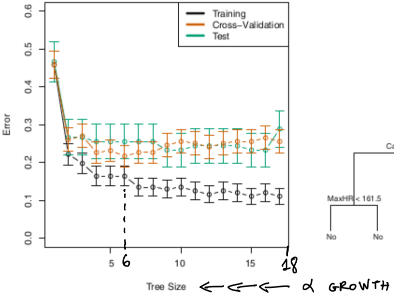


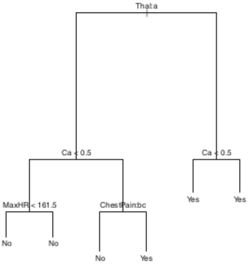
$$\sum_{p=1}^{|T|} \Big(- \sum_{k=1,K} f_k \log(f_k) \Big) + \alpha |T|$$

  3. operate a cross-validation to determine the best α value and so the best sub-tree

a pruned classification tree







some decision trees considerations

- Gini coefficient can be used as a impurity measure
- in the regression version the predicted target of leaf nodes is the mean value of training observations falling in that node
- decision trees are close to human decision-making mechanisms
- are easy to depict and interpret
- can deal with mixed features
- no high level of accuracy compared to other models
- not robust to data → DATA NOISE, AND OUTLIERS

⇒ to overcome the previous drawbacks multiple trees are combined together (random forests)

ENSEMBLE MODELS

exercise 4.4

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Support Vector Machines

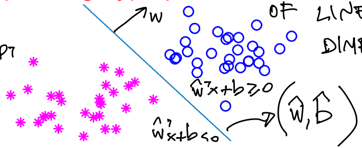
SVMs are machine learning models for binary classification (adaptable to multiclass and regression)

- binary classification problem

$$TR = \{(x^i, y^i) : x^i \in \Re^n, y^i \in \{-1, +1\}, i = 1, \ldots, m\}$$

- assume TR is linearly separable, i.e. the points of the two classes can be EXACTLY separated by a $(n-1)$-dimensional separating hyperplane → GENERALIZATION OF LINE AT DIMENSION = n

$$w^T x + b = 0, \quad \underbrace{w}_{\text{NORMAL VECTOR}} \in \Re^n, \underbrace{b}_{\text{INTERCEPT}} \in \Re$$



- training a SVM ⇒ determine optimal parameters $(\hat{w}, \hat{b})$ of the hyperplane separating samples in TR according to their class
- once trained the model, the decision function $h(\cdot)$ to classify unseen observations $\tilde{x}$ is

$$h(\tilde{x}) = sign(\hat{w}^T \tilde{x} + \hat{b})$$

IF $\hat{w}^T \tilde{x} + \hat{b} \geq 0 \rightarrow$ ○

IF $\hat{w}^T \tilde{x} + \hat{b} < 0 \rightarrow$ *

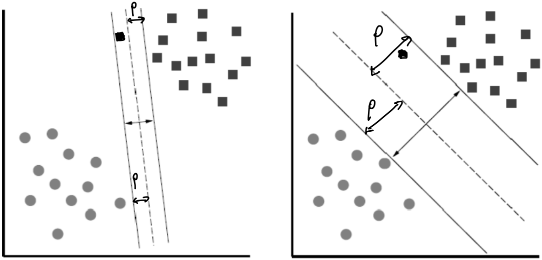
- if TR is linearly separable, a separating hyperplane (w, b) divides data with a tolerance gap ρ according to the separating conditions

SEPARATION WITH MARGIN $\rho > 0$

$$\begin{cases} w^T x^i + b \geq \rho & \text{for } y^i = +1 \\ w^T x^j + b \leq -\rho & \text{for } y^j = -1 \end{cases} \quad \overset{\text{NORMALIZED VERSION}}{\Longrightarrow} \quad \begin{cases} w^T x^i + b \geq 1 & \text{for } y^i = +1 \\ w^T x^j + b \leq -1 & \text{for } y^j = -1 \end{cases}$$

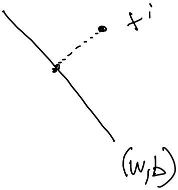
SIMPLE SEPARATION:
$$\begin{cases} w^T x^i + b \geq 0 & \forall y^i = 1 \\ w^T x^i + b < 0 & \forall y^i = -1 \end{cases}$$

- the separating hyperplane are infinitely many! which one is better?



- the maximum margin hyperplane (the one with maximum distance from the closest points) is better!
  - hyperplanes close to training samples are more sensitive to noise
  - intuitively/theoretically a big margin hyperplane is more likely to better classifiy unseen data

a maximum margin training problem...



- the distance of a point x^i from hyperplane (w, b) is

$$d(x^i; (w, b)) = \frac{|w^T x^i + b|}{\|w\|}$$

SCALAR

SCALAR

- the margin of a hyperplane (w, b) for TR is

$$\rho(w, b) = \min_{i=1,\dots,m} \left\{ \frac{|w^T x^i + b|}{\|w\|} \right\}$$

- it can be shown that: maximizing $\min_{i=1,\dots,m} \left\{ \frac{|w^T x^i + b|}{\|w\|} \right\} \equiv$ minimizing $\|w\|$
- SVM training optimization problem

UNIQUE

MUST BE MIDWAY BETWEEN THE 2 CLASSES

MAXIMIZATION OF THE MARGIN

$$\min_{w \in \Re^n, b \in \Re} \quad \frac{\|w\|^2}{2}$$

DIFFERENTIABLE

$$\text{subject to} \quad \underbrace{y^i(w^T x^i + b) \geq 1}_{\text{SEPARATING CONDITIONS WITH MARGIN}} \quad i = 1, \dots, m$$

- it is a convex problem!
  - EFFICIENTLY SOLVED
  - UNIQUE SOLUTION